

Disaster Recovery

In this section we will test the Disaster Recovery (DR) procedures for automatic failover and manual failback. We will perform this in two phases – **Failover** and **Failback** – and each phase will be broken down into multiple steps for, we hope, clarity.

The first phase will include the following steps:

- 1) First, we will enable the DR user.
- 2) Then, start the VM **drvault01**.
- 3) Then, we will configure the **Disaster Recovery module** on the **DR** server to perform an automatic failover in case the **Primary Vault** is no longer reachable.
- 4) We will execute a full replication from the **Primary Vault** to the **DR Vault**.
- 5) We will perform an automatic failover from the **Primary Vault** to the **DR Vault**. As part of the test, we will also confirm that our end users can still access critical systems via **CyberArk**, without any human intervention.

Once the failover is completed, we will return our systems to their original state by performing what we call a Failback.

- 1) We will set the **Primary Vault** to act as DR and replicate all data back from the **DR Vault** to the **Primary Vault**.
- 2) We will then perform a manual failback from the **DR Vault** to the **Primary Vault**.
- 3) Lastly, we will set the **DR Vault** back to DR mode and confirm our end users are still able to connect to critical systems via **CyberArk**.

The steps below have already been performed by the implementation team:

- The **PrivateArk Server**, **PrivateArk Client**, and **Disaster Recovery** module have all been installed on both your **vault01** and DR servers by the implementation team.
- A second **DR** user called “**DR_Failback**” was manually created by the implementation team during the deployment of the **Primary Vault** for the purpose of supporting the failback procedure from the DR site back to the primary site.

Step 1: Enable DR user

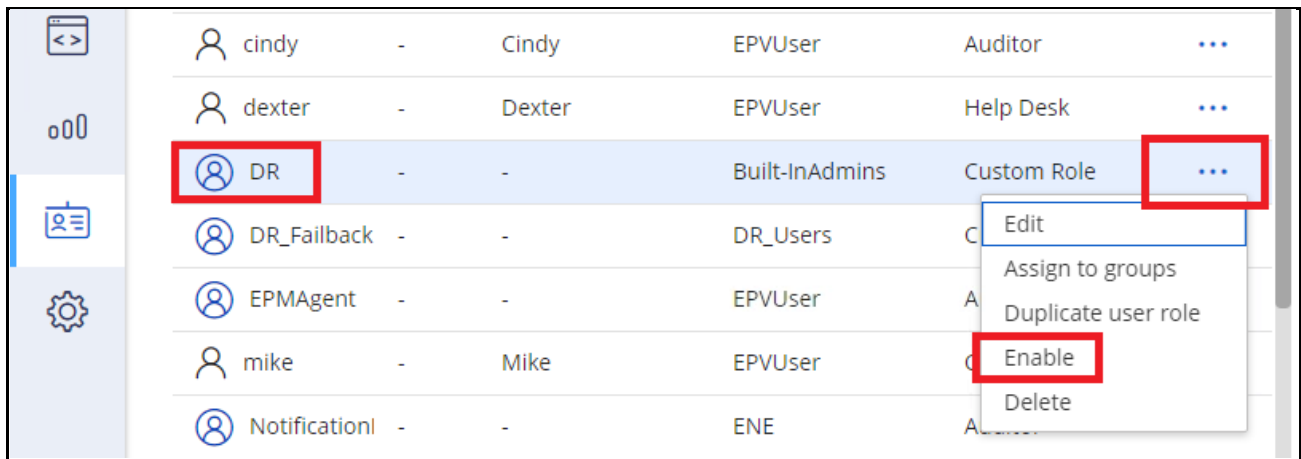
As noted above, the implementation team has already installed the **PrivateArk Server**, **PrivateArk Client** and **Disaster Recovery** service on the **DR** server. However, to avoid an unwanted automatic failover during the first days of the course, automatic failover was disabled, and the DR user deactivated. We are now going to enable the DR user on the **Primary Vault** and Automatic Failover to the **DR Vault**.

1. Launch Chrome to open the **PVWA** page. Log in as *Mike*. Go to the **User Provisioning** tab and then **Users**.

	Policies	▼
	Security	▼
	Applications	
	Reports	
	User Provisioning	^
	Users	
	Groups	
	Administration	▼

	Auditor	-	-	Built-InAdmins	Auditor
	Backup	-	-	Built-InAdmins	Custom Role
	carlos	-	Carlos	EPVUser	Standard User
	cindy	-	Cindy	EPVUser	Auditor
	dexter	-	Dexter	EPVUser	Help Desk
	DR	-	-	Built-InAdmins	Custom Role
	DR_Fallback	-	-	DR_Users	Custom Role
	EPMAgent	-	-	EPVUser	Auditor
	mike	-	Mike	EPVUser	Custom Role
	NotificationEngine	-	-	ENE	Auditor
	Operator	-	-	Built-InAdmins	Custom Role
	PasswordManager	-	-	CPM	Standard User

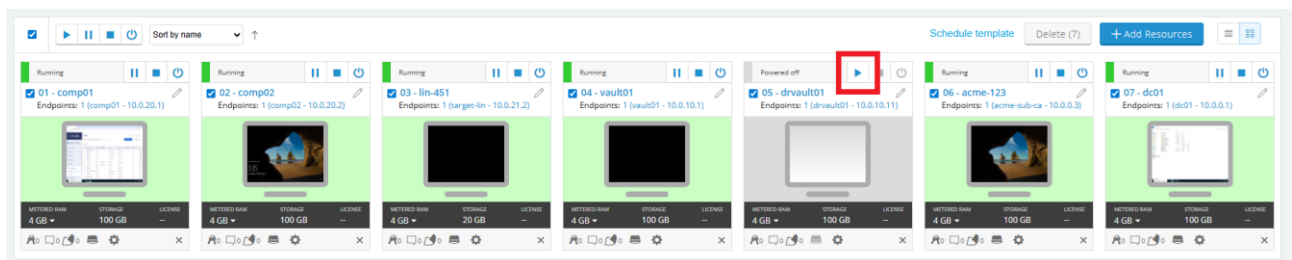
2. Locate the *DR* user, click on the ellipsis and **Enable**.



Remember! Do **NOT** reset the *DR* user's password.

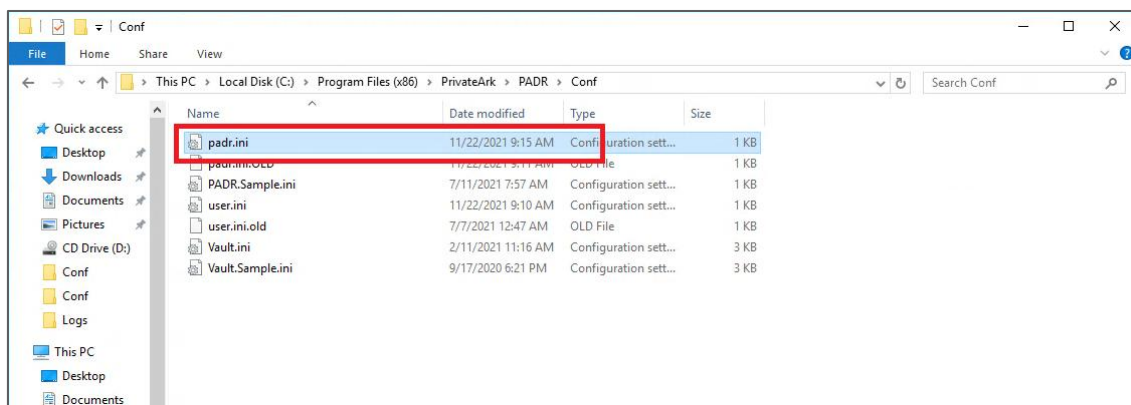
Step 2: Start the DR Vault

- Next, power on the **DR** server. Remember, it will take a moment for the machine to start.

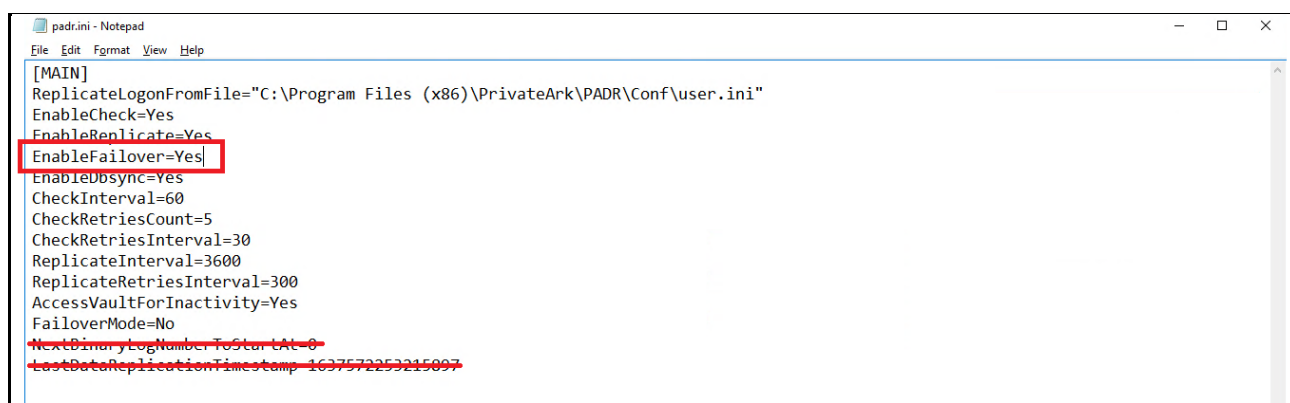


Step 3: Activate Automatic Failover

- Sign into Windows on the **drvault01** server as *Administrator* with the password *Cyberark1*.
- Open the file explorer and navigate to *C:\Program Files (x86)\PrivateArk\PADR\Conf*. Double click on the *padr.ini* file to edit it with Notepad (there is a short-cut to the *PADR* folder on the desktop).



- Verify that the *EnableFailover* setting is set to **Yes** and delete the last two lines of the file (if present). This will trigger a full replication when we restart the Disaster Recovery service, ensuring we have the most up-to-date data.



Note: Notice *FailoverMode* is currently set to **No**. Do NOT change this setting. It will automatically change later when we test the failover process.

- Save the file and exit Notepad.

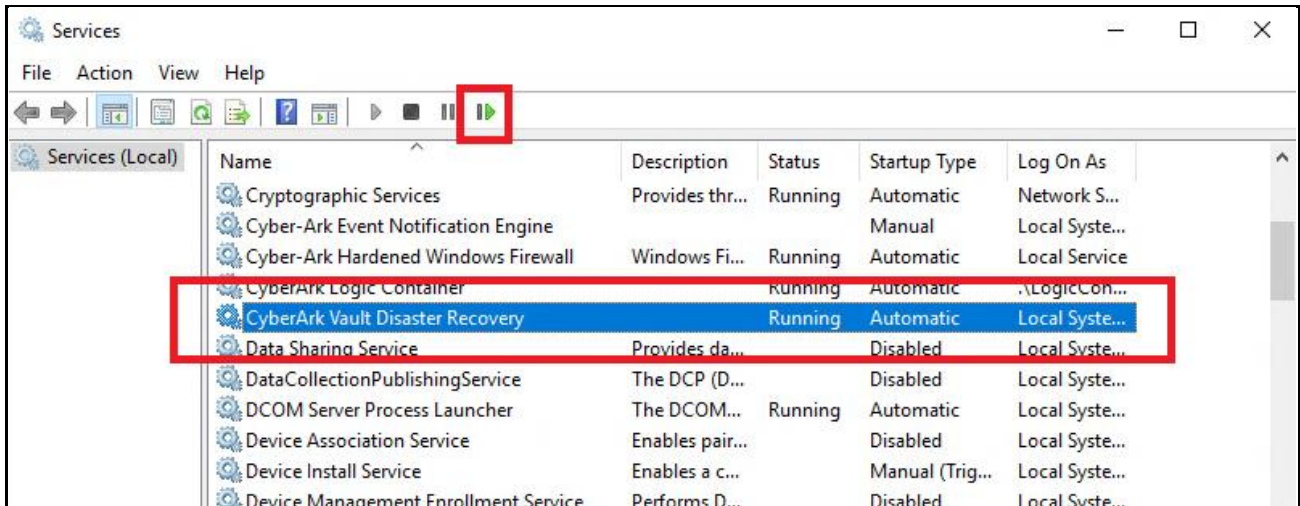
Step 4: Execute a full replication to the DR Vault

In this step, we will simply re-start the **CyberArk Disaster Recovery** service. In so doing, the service will read the changes we made to the file *PADR.ini*.

- On the **DR** server, open the Windows Services applet. There is a shortcut in the task bar.



2. Restart the **CyberArk Vault Disaster Recovery** service.



3. Now go to the desktop. Right click on the ***padr_log.ps1*** file located on the desktop and select **Run with PowerShell**.

Note: The above script will run a tail on the *padr.log* file located in *C:\Program Files (x86)\PrivateArk\PADR\logs* folder. The tail will allow you to monitor the actions performed by the Disaster Recovery service in real time. If you are prompted to allow running the script, select **Yes**.

4. Confirm the Disaster Recovery module has completed the replication of data from the **Primary Vault**. You should see entries with informational codes **PAREP013I** **Replicating Safe** and at the end:

- **PADR0010I Replicate ended**
- **PADR0099I Metadata Replication is running successfully.**

```
[22/11/2021 09:21:27.458853] :: PAREP013I Replicating Safe PSMPADBridgeCustom.
[22/11/2021 09:21:27.473839] :: PAREP013I Replicating Safe TEST.
[22/11/2021 09:21:27.486196] :: PAREP013I Replicating Safe CyberArk-Service-Accounts.
[22/11/2021 09:21:27.502349] :: PAREP013I Replicating Safe Lin-Fin-US.
[22/11/2021 09:21:27.518590] :: PAREP013I Replicating Safe Ora-Fin-US.
[22/11/2021 09:21:27.534876] :: PAREP013I Replicating Safe Win-Srv-Fin-US.
[22/11/2021 09:21:27.706771] :: PAREP013I Replicating Safe TEST-RESTORE.
[22/11/2021 09:21:27.944510] :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
[22/11/2021 09:21:27.944597] :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
[22/11/2021 09:21:28.199807] :: PADR0010I Replicate ended.
[22/11/2021 09:21:29.217971] :: PADR0099I Metadata Replication is running successfully.
```

Note: Keep the tail running for the remainder of the exercise.

- Back on the **comp01** server and login to the **PVWA** as **Mike**.
- Navigate to **System Health** to review the status of the DR Vault. Note that the Primary Vault is online and the **DR on 10.0.10.5** is functioning normally.

System Health

Last sign in: 2/1/2026 |  mike ▾

Updated at: 3:52 PM 

Vaults



PRIMARY

» DR 10.0.10.1 

» DR 10.0.10.5 

-  DR user is connected to the Primary
-  DR database replication is up to date
-  DR files replication is up to date

Note: The **System Health** is also aware that there is a Disaster Recovery component on **10.0.10.1**, that is our Primary Vault, but that it is currently offline. This is normal.

Step 5: Test Automatic Failover

Now, we will execute an automatic failover test by stopping the **PrivateArk Server** service on the **Primary Vault**. Perform this action using the **Remote Control Client** as described in an earlier exercise using the command “stop vault”.



```
Windows PowerShell
PS C:\Remote Control Client> .\PARClient.exe 10.0.10.1

Cyber-Ark Remote Administration Client (15.0.0.76)

Working with agent on: 10.0.10.1

Loaded component from [C:\Remote Control Client\PARClusterVaultClient.dll]
Loaded component from [C:\Remote Control Client\PARDRClient.dll]
Loaded component from [C:\Remote Control Client\PARNEClient.dll]
Loaded component from [C:\Remote Control Client\PARVaultClient.dll]

PARCLIENT> stop vault
Are you sure you want to stop the remote Vault (Y/N)? y
Password: *****
Vault was stopped successfully
PARCLIENT> status vault
Vault is stopped.

PARCLIENT>
```

- Once the **Primary Vault** has stopped, return to the console of the **DR Server**.
- Monitor the the tail on the *padr.log* file. You should see messages stating that the Disaster Recovery service is unable to reach the **Primary Vault**.

```
Windows PowerShell
[22/11/2021 13:45:46.617139] :: PAREP013I Replicating Safe Ora-Fin-US.
[22/11/2021 13:45:46.632168] :: PAREP013I Replicating Safe Win-Srv-Fin-US.
[22/11/2021 13:45:46.646869] :: PAREP013I Replicating Safe TEST-RESTORE.
[22/11/2021 13:45:46.670555] :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
[22/11/2021 13:45:46.670583] :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
[22/11/2021 13:45:46.678500] :: PADR0010I Replicate ended.
[22/11/2021 13:45:47.682944] :: PADR0099I Metadata Replication is running successfully.
[22/11/2021 13:48:20.590632] :: PADR0005E CASTM003E Vault transaction failed. Reason: ITACM012S Timeout has expi
red. . (Code 24)
[22/11/2021 13:48:20.590815] :: PADR0014E Attempt to test vault availability failed (code=1).
```

Note: If you are not seeing new entries in the log file after a few minutes, press **Enter**. If you are still not seeing new entries, close the PowerShell window and run the script again.

- After 5 failures, the **DR Vault** will go into failover mode (this is the default setting). Check the *padr.log* and review the sequence of events.

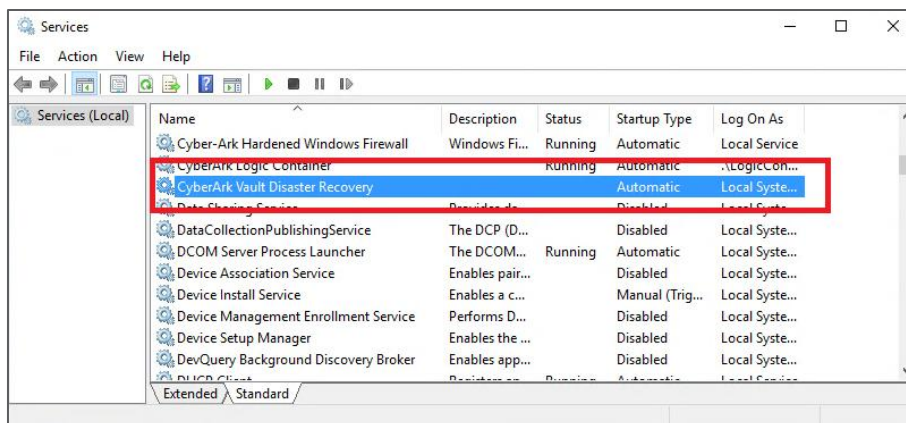
```

Windows PowerShell
22/11/2021 13:45:46.584614 :: PAREP013I Replicating Safe CyberArk-Service-Accounts.
22/11/2021 13:45:46.600744 :: PAREP013I Replicating Safe Lin-Fin-US.
22/11/2021 13:45:46.617139 :: PAREP013I Replicating Safe Ora-Fin-US.
22/11/2021 13:45:46.632168 :: PAREP013I Replicating Safe Win-Srv-Fin-US.
22/11/2021 13:45:46.646869 :: PAREP013I Replicating Safe TEST-RESTORE.
22/11/2021 13:45:46.670555 :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
22/11/2021 13:45:46.670583 :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
22/11/2021 13:45:46.678500 :: PADR0010I Replicate ended.
22/11/2021 13:45:47.682944 :: PADR0099I Metadata replication is running successfully.
22/11/2021 13:48:20.590632 :: PADR0005E CASTM003E Vault transaction failed. Reason: ITACM012S Timeout has expired. . (Code 24)
22/11/2021 13:48:20.590815 :: PADR0014E Attempt to test vault availability failed (code=1).
22/11/2021 13:49:20.026424 :: PADR0005E CASTM003E Vault transaction failed. Reason: ITACM062S Communication error (Diagnostic information: 562) . (Code 24)
22/11/2021 13:49:20.026466 :: PADR0015E Attempt to test vault availability failed 2 times (code=-1066062).
22/11/2021 13:50:20.480362 :: PADR0005E CASTM003E Vault transaction failed. Reason: ITACM062S Communication error (Diagnostic information: 562) . (Code 24)
22/11/2021 13:50:20.480415 :: PADR0015E Attempt to test vault availability failed 3 times (code=-1066062).
22/11/2021 13:50:47.886213 :: PADR0099I Metadata Replication is running successfully.
22/11/2021 13:51:20.948173 :: PADR0005E CASTM003E Vault transaction failed. Reason: ITACM062S Communication error (Diagnostic information: 562) . (Code 24)
22/11/2021 13:51:20.948214 :: PADR0015E Attempt to test vault availability failed 4 times (code=-1066062).
22/11/2021 13:52:20.354379 :: PADR0005E CASTM003E Vault transaction failed. Reason: ITACM062S Communication error (Diagnostic information: 562) . (Code 24)
22/11/2021 13:52:20.354426 :: PADR0015E Attempt to test vault availability failed 5 times (code=-1066062).
22/11/2021 13:52:20.354963 :: PADR0016E Vault availability test failed, failover started.
22/11/2021 13:52:20.355290 :: CyberArk Failover method
22/11/2021 13:52:20.355317 :: Performing Active/Passive fail-over procedure: PADRDoActivePassiveFailover
22/11/2021 13:52:20.368678 :: PADR0103I Failover process started.
22/11/2021 13:52:20.368704 :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
22/11/2021 13:52:20.379025 :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
22/11/2021 13:52:20.394734 :: ITATS408I Synchronizing objects of Safe Notification Engine...
22/11/2021 13:52:20.399671 :: ITATS408I Synchronizing objects of Safe PVWATaskDefinitions...
22/11/2021 13:52:20.401693 :: ITATS158I Deleting total of 0 objects.
22/11/2021 13:52:31.449946 :: PADR0025I Failover process ended successfully.
22/11/2021 13:52:31.449997 :: PADR0067I Starting Vault service.
22/11/2021 13:52:41.121832 :: PADR0017I Failover completed, PADR service is shutting down.
22/11/2021 13:52:41.167005 :: PADR0022I Disaster Recovery service terminated.
  
```

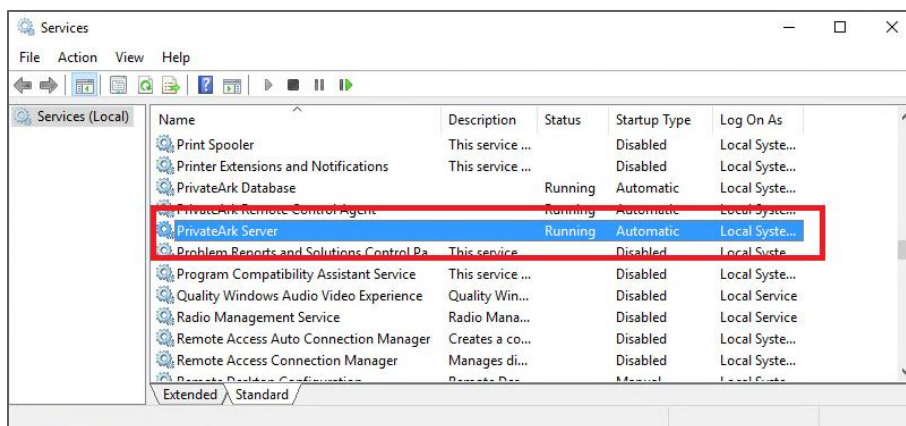
Note: The entire process should take around 5 minutes.

Confirm Automatic Failover on the DR Vault

- On the **DR server (10.0.10.5)**, open the Windows Services applet and confirm the **CyberArk Vault Disaster Recovery** service has terminated.



- Confirm the **PrivateArk Server** service is now running on the **DR server (10.0.10.5)**.

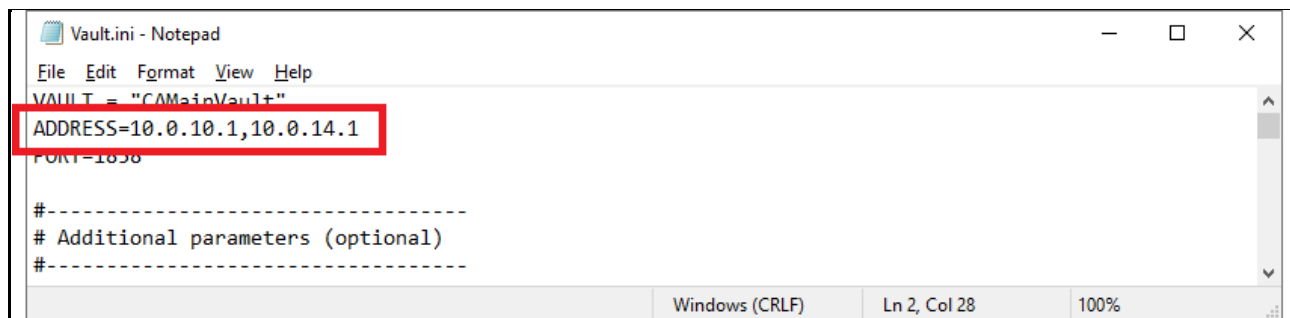


Confirm Automatic Failover of PVWA and PSM

In this section we will confirm that our end users (like *John*) can still access critical systems via **CyberArk**, even though the **Primary Vault** is offline, without human intervention.

Note the implementation team has already configured the **PVWA** and **PSM** to automatically failover to the **DR Vault** when the **Primary Vault** is no longer available. To support automatic failover, the *Vault.ini* file for both services has been configured with the IP addresses of both the **Primary Vault** and the **DR Vault** separated by a comma. The **CPM** has not been configured for automatic failover, nor should it be.

Here you can see the configuration of the **PVWA** *Vault.ini* file:



1. To confirm that both the **PVWA** and **PSM** automatic failover was successful, return to the console of the **comp01** server.
2. Open Chrome and verify that you can still login to the **PVWA** as *John*, even though the **Primary Vault** is offline.

3. Now, verify you can launch a secure session to the target Windows machine using the *administrator* account via **PSM**. If everything worked as expected, *John* should still be able to access the target server via **CyberArk**, without any human intervention.

Note: You may need to try to launch the connection via **PSM** a couple of time before it works, as it may take a few minutes before the **PSM** fails over to the **DR Vault**.

Failback

Now let's say that the problem with our old Primary Vault – **vault01** – has been resolved and we want to return it to its role as the Primary Vault. During the time of the failover, however, the DR Vault was acting as the Primary and recording the activities of users (although no passwords have been changed because we did not failover the **CPM**). So, before we failback to the **Primary Vault** we must first make sure we replicate all the latest data from the **DR Vault** (which served as the active **Vault** for the duration of resolving the incident). In this section we will use the **Disaster Recovery** module on **vault01** to replicate data back from the **DR Vault** to the **Primary Vault**.

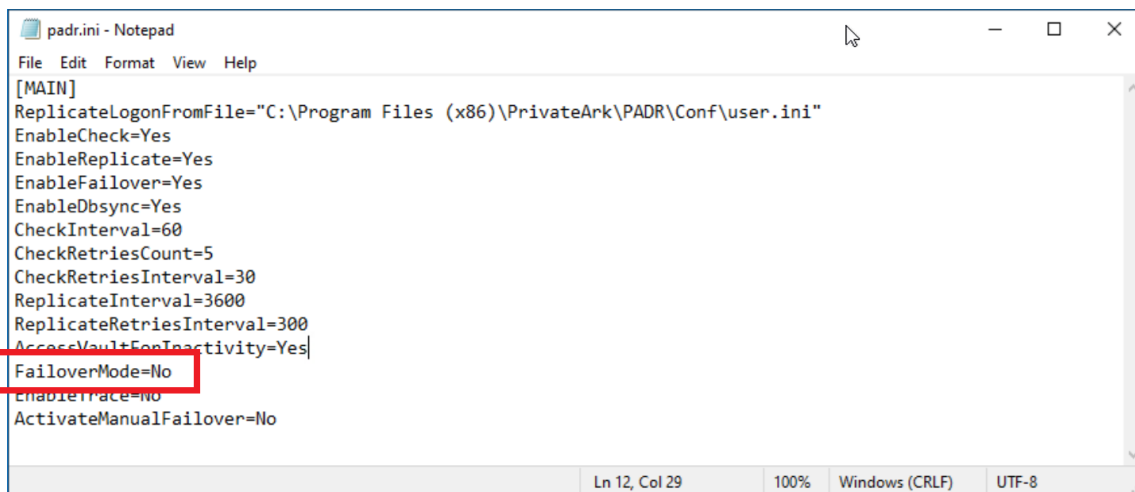
Note: The implementation team has already installed the **Disaster Recovery** module on **vault01**, and manually created a separate DR user for the purpose of performing replication from the **DR Vault** back to the **Primary Vault**.
The new user is called *DR_Failback*, and has been made a member of the built-in group *DR_Users*. The user was assigned the following **Vault** authorizations: *Backup All Safes* and *Restore All Safes*.

Step 1: Execute a full replication back to the Primary Vault

1. First, we need to enable the *DR_Failback* user for the **DR Vault**. You can do this either in the **PrivateArk Client** or the **PVWA**.
2. We have already seen how to enable a user in the **PVWA**. To do it from the **PrivateArk Client** (which until recently was the only way it could be done), from **Comp01**, open the **PrivateArk Client**. Double-click on the **DR Vault** shortcut and login as *Administrator*. To enable the user *DR_Failback* by going to **Tools | Administrative Tools | Users and Groups**. Scroll down the list to the **System** location. Click on *DR_Failback*, then **Update**, then uncheck the box for **Disable user**.

Note: The process is identical to [enabling the Auditor user](#) performed earlier. Do **NOT** change the password as the **DR_Failback** user has already authenticated to the **Vault** during initial implementation, and the password for the user has already been rotated. If you change the password to *Cyberark1*, you will need to create a new cred file as well.

3. Next, we to configure Disaster Recovery on the **Primary Vault**, so connect to the VM **VAULT01 (10.0.10.1)**.
4. Open the file explorer and navigate to *C:\Program Files (x86)\PrivateArk\PADR\Conf*. Double click on the *padr.ini* file to edit it with Notepad.
5. Make the following changes to the *padr.ini* file on **vault01**:
 - Set *FailoverMode* to **No** if it is not set yet.
 - If the parameters ***NextBinaryLogNumberToStartAt=-1*** and ***LastDataReplicationTimestamp*** appear in the file, delete them.

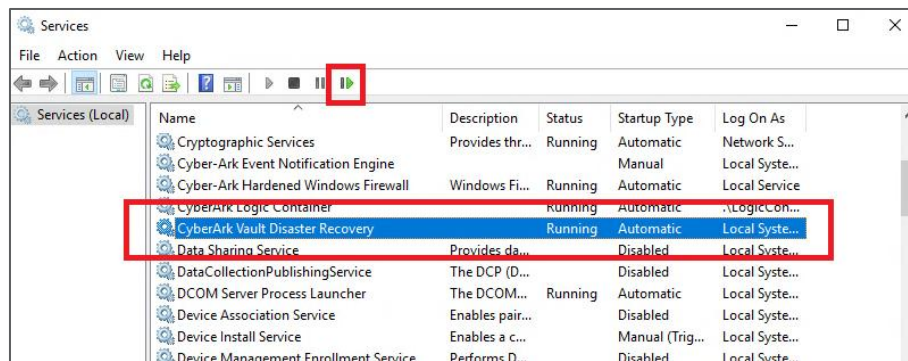


```

padr.ini - Notepad
File Edit Format View Help
[MAIN]
ReplicateLogonFromFile="C:\Program Files (x86)\PrivateArk\PADR\Conf\user.ini"
EnableCheck=Yes
EnableReplicate=Yes
EnableFailover=Yes
EnableObsync=Yes
CheckInterval=60
CheckRetriesCount=5
CheckRetriesInterval=30
ReplicateInterval=3600
ReplicateRetriesInterval=300
AccessVaultOnInactivity=Yes
FailoverMode=No
EnableTrace=No
ActivateManualFailover=No
Ln 12, Col 29 100% Windows (CRLF) UTF-8
  
```

Note: The above changes will trigger the Disaster Recovery module on the Primary Vault to perform a full replication of the data from the DR Vault once the service is restarted.

6. Save the file and close it.
7. Start the **CyberArk Vault Disaster Recovery** service on the **Primary Vault**.



8. Once the Disaster Recovery service is running, Right click on the *padr_log.ps1* file located on the desktop of the **vault01** and select **Run with PowerShell**.

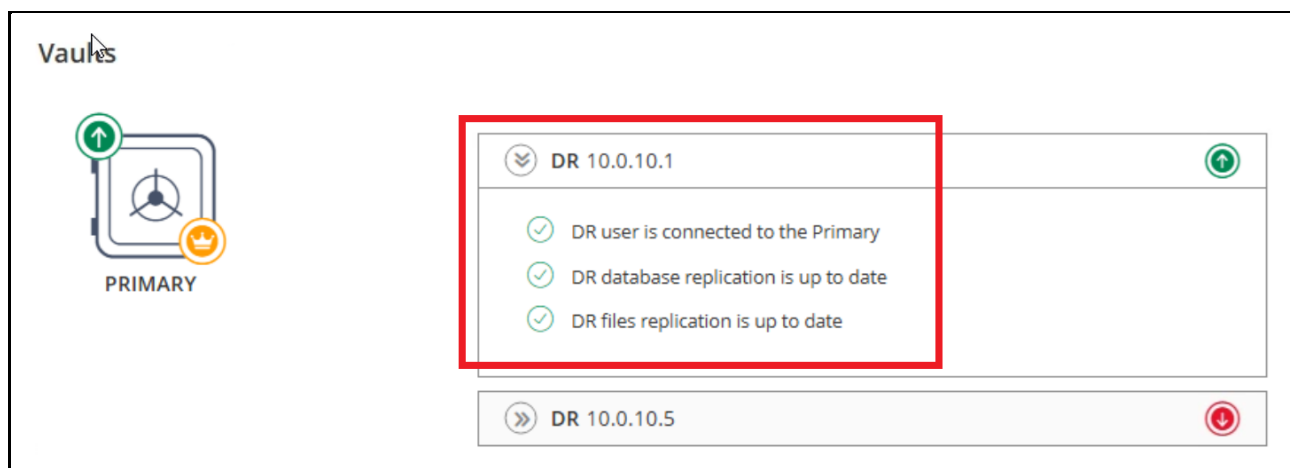
Note: If you are prompted to allow running the script, select **Yes**.

9. Monitor the tail of the *padr_log.ps1* to verify that the **Primary Vault** has replicated all the changes from the **DR Vault**.

```

Select Windows PowerShell
[22/11/2021 14:13:35.752086] :: PAREP013I Replicating Safe PSMSessions.
[22/11/2021 14:13:35.787257] :: PAREP013I Replicating Safe PSMLiveSessions.
[22/11/2021 14:13:35.798238] :: PAREP013I Replicating Safe PSMUniversalConnectors.
[22/11/2021 14:13:35.812530] :: PAREP013I Replicating Safe PSMNotifications.
[22/11/2021 14:13:35.828253] :: PAREP013I Replicating Safe PSMUnmanagedSessionAccounts.
[22/11/2021 14:13:35.848200] :: PAREP013I Replicating Safe Transfer.
[22/11/2021 14:13:35.859998] :: PAREP013I Replicating Safe PSMRecordings.
[22/11/2021 14:13:35.948798] :: PAREP013I Replicating Safe PSMPConf.
[22/11/2021 14:13:35.955371] :: PAREP013I Replicating Safe PSMPLiveSessions.
[22/11/2021 14:13:35.969742] :: PAREP013I Replicating Safe PSMPADBridgeConf.
[22/11/2021 14:13:35.984625] :: PAREP013I Replicating Safe PSMPADBridgeUserProfile.
[22/11/2021 14:13:36.005684] :: PAREP013I Replicating Safe PSMPADBridgeCustom.
[22/11/2021 14:13:36.015617] :: PAREP013I Replicating Safe TEST.
[22/11/2021 14:13:36.032397] :: PAREP013I Replicating Safe CyberArk-Service-Accounts.
[22/11/2021 14:13:36.048438] :: PAREP013I Replicating Safe Lin-Fin-US.
[22/11/2021 14:13:36.062996] :: PAREP013I Replicating Safe Ora-Fin-US.
[22/11/2021 14:13:36.078828] :: PAREP013I Replicating Safe Win-Srv-Fin-US.
[22/11/2021 14:13:36.094645] :: PAREP013I Replicating Safe TEST-RESTORE.
[22/11/2021 14:13:36.114298] :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
[22/11/2021 14:13:36.114333] :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
[22/11/2021 14:13:36.174906] :: PADR0010I Replicate ended.
[22/11/2021 14:13:37.177795] :: PADR0099I Metadata Replication is running successfully.
  
```

10. On the **comp01** server, login to the **PVWA** as **Mike**. Navigate to **SYSTEM HEALTH** to review the current system health. Note that Vault **10.0.10.1** is now the **DR**, while the old DR – **10.0.10.5** – is showing as disconnected, which it is with respect to its former role as the DR Vault.



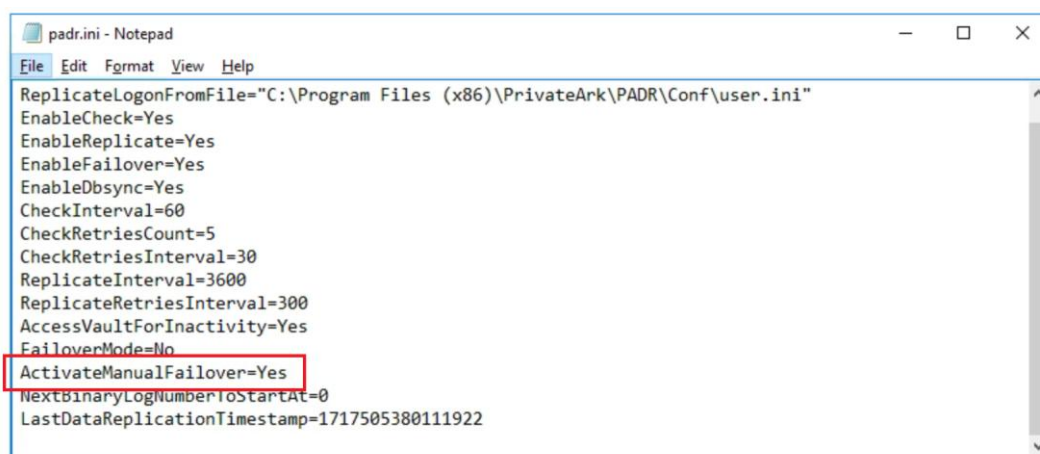
Step 2: Stop the PrivateArk Server service on drvault01

To prevent there being two Vaults active at a given moment, first stop the PrivateArk Server service on **drvault01** (10.0.10.5).

Step 3: Perform a manual failback procedure

Now we can proceed with performing a manual failback from the **DR Vault** to the **Primary Vault**. The failback procedure will be performed using a **Manual Failover**.

1. Make sure you are working on **vault01** (10.0.10.1).
2. Open the file explorer and navigate to *C:\Program Files (x86)\PrivateArk\PADR\Conf*. Double click on the *padr.ini* file to edit it with Notepad.
3. Set *ActivateManualFailover* to **Yes**.



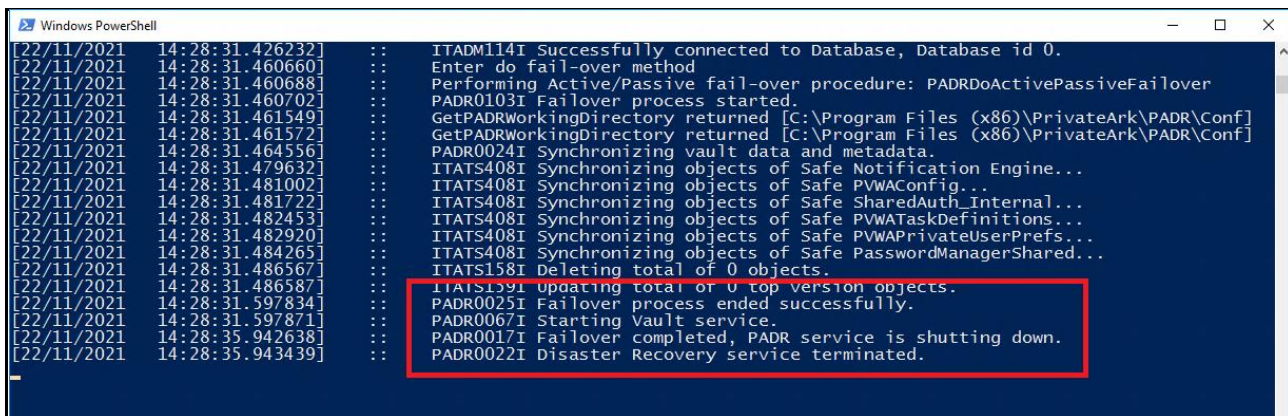
4. Save the file and close it.

5. Start the **CyberArk Vault Disaster Recovery** service on **vault01** (10.0.10.1). The service should start and stop immediately (because of the *ActivateManualFailover* setting). Then the **PrivateArk Server** service should start.

Important: The above steps are critical for a successful failback from the DR Vault to the Primary Vault. Reverting to the Primary Vault without first performing a proper failover can result in data inconsistencies.

Confirm Manual Failover on the Primary Vault

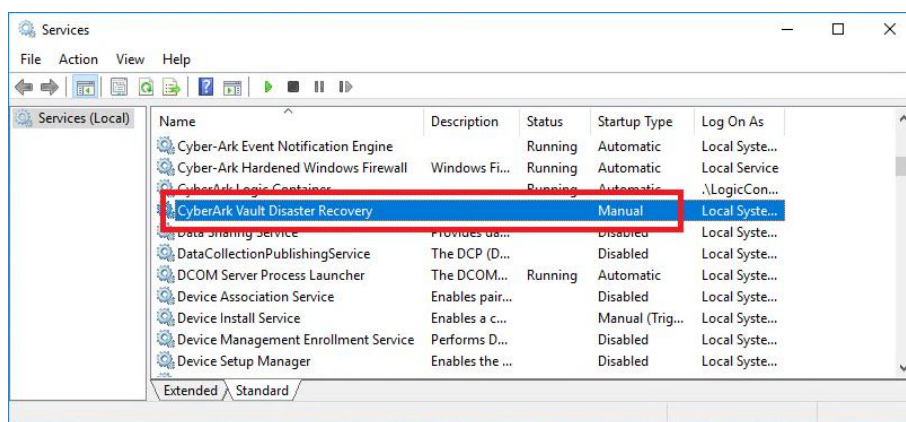
1. Monitor the tail running on the *padr.log* file on **vault01** (10.0.10.1). Confirm you can see the messages stating that the Failover process ended successfully, that the Vault service is starting, and that the *Disaster Recovery* service has terminated.



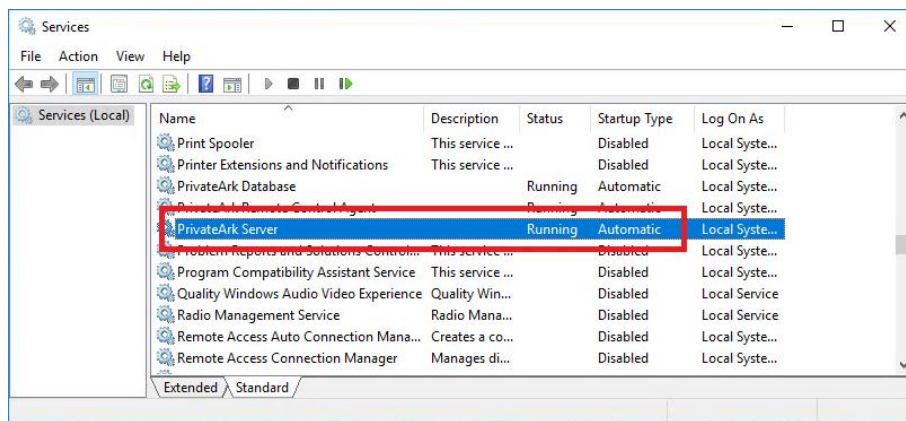
```

[22/11/2021 14:28:31.426232] :: ITADM114I Successfully connected to Database, Database id 0.
[22/11/2021 14:28:31.460660] :: Enter do fail-over method
[22/11/2021 14:28:31.460688] :: Performing Active/Passive fail-over procedure: PADRDoActivePassiveFailover
[22/11/2021 14:28:31.460702] :: PADR0103I Failover process started.
[22/11/2021 14:28:31.461549] :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
[22/11/2021 14:28:31.461572] :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
[22/11/2021 14:28:31.464556] :: PADR0024I Synchronizing vault data and metadata.
[22/11/2021 14:28:31.479632] :: ITATS408I Synchronizing objects of Safe Notification Engine...
[22/11/2021 14:28:31.481002] :: ITATS408I Synchronizing objects of Safe PVWAConfig...
[22/11/2021 14:28:31.481722] :: ITATS408I Synchronizing objects of Safe SharedAuth Internal...
[22/11/2021 14:28:31.482453] :: ITATS408I Synchronizing objects of Safe PVWATaskDefinitions...
[22/11/2021 14:28:31.482920] :: ITATS408I Synchronizing objects of Safe PVWAPrivateUserPrefs...
[22/11/2021 14:28:31.484265] :: ITATS408I Synchronizing objects of Safe PasswordManagerShared...
[22/11/2021 14:28:31.486567] :: ITATS158I Deleting total of 0 objects.
[22/11/2021 14:28:31.486587] :: ITATS159I Updating total of 0 top version objects.
[22/11/2021 14:28:31.597834] :: PADR0025I Failover process ended successfully.
[22/11/2021 14:28:31.597871] :: PADR0067I Starting Vault service.
[22/11/2021 14:28:35.942638] :: PADR0017I Failover completed, PADR service is shutting down.
[22/11/2021 14:28:35.943439] :: PADR0022I Disaster Recovery service terminated.
  
```

2. Verify that the the **CyberArk Vault Disaster Recovery** service has terminated on **vault01** (10.0.10.1).



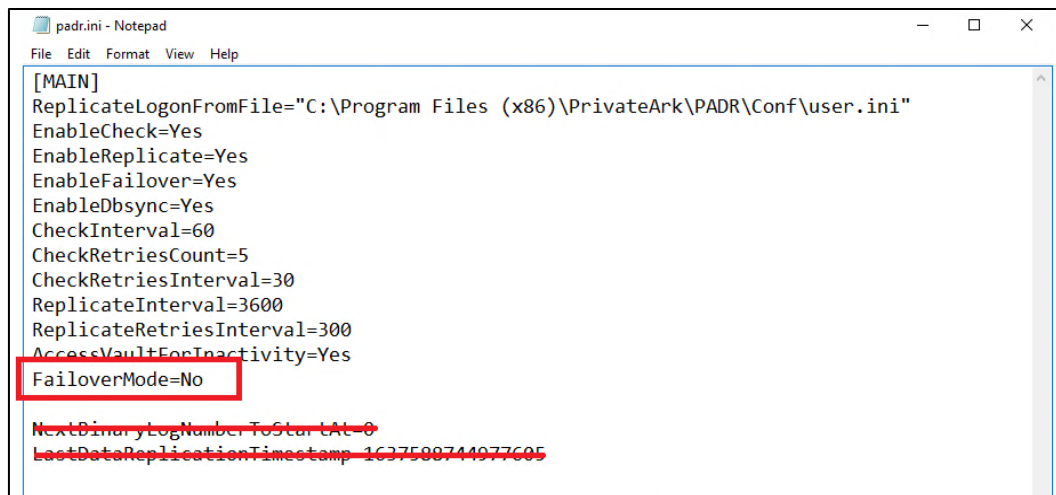
3. Verify that the **PrivateArk Server** service has started successfully on **vault01** (10.0.10.1).



Step 4: Set the DR server back to DR mode

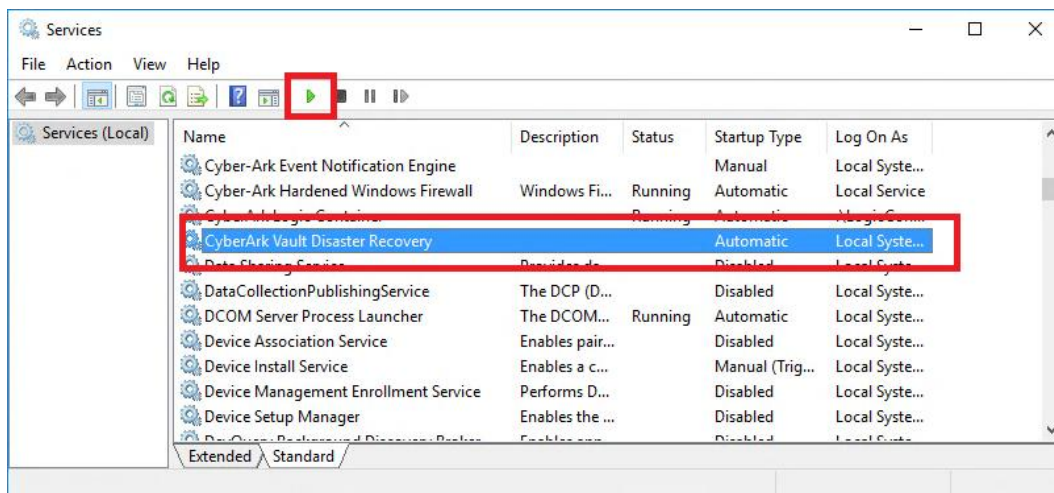
In the last section of this exercise, we will set the **DR** server back to DR mode.

1. Return to the console of **DR** (10.0.10.5).
2. On the **DR** server, edit the *padr.ini* file and make the following changes:
 - Set *FailoverMode* to **No**.
 - Delete the last two lines (log number and timestamp of the last successful replication) in the file.
 - Save and exit the file.



Note: Click **Yes** to stop the **Cyber-Ark Event Notification Engine** service as well.

3. Then, **start the CyberArk Vault Disaster Recovery** service.



4. Check the tail running on the *padr.log* file on the **DR** server (10.0.10.5) and confirm that a full replication process started and that the replication (from the **Primary Vault** to the **DR Vault**) has ended successfully.

```

Windows PowerShell
22/11/2021 14:35:02.9418541 :: PAREP013I Replicating Safe PVWATicketingSystem.
22/11/2021 14:35:02.9579511 :: PAREP013I Replicating Safe PVWAPublicData.
22/11/2021 14:35:02.9741457 :: PAREP013I Replicating Safe PVWATaskDefinitions.
22/11/2021 14:35:02.9892507 :: PAREP013I Replicating Safe PVWAPrivateUserPrefs.
22/11/2021 14:35:03.0051877 :: PAREP013I Replicating Safe PasswordManager.
22/11/2021 14:35:03.0196757 :: PAREP013I Replicating Safe PasswordManager_workspace.
22/11/2021 14:35:03.0355187 :: PAREP013I Replicating Safe PasswordManager_ADInternal.
22/11/2021 14:35:03.0515687 :: PAREP013I Replicating Safe PasswordManager_Info.
22/11/2021 14:35:03.0684867 :: PAREP013I Replicating Safe PasswordManagerShared.
22/11/2021 14:35:03.0869327 :: PAREP013I Replicating Safe PasswordManagerTemp.
22/11/2021 14:35:03.0985947 :: PAREP013I Replicating Safe PasswordManager_Pending.
22/11/2021 14:35:03.1143397 :: PAREP013I Replicating Safe AccountsFeedADAccounts.
22/11/2021 14:35:03.1320367 :: PAREP013I Replicating Safe AccountsFeedDiscoveryLogs.
22/11/2021 14:35:03.1466727 :: PAREP013I Replicating Safe PSM.
22/11/2021 14:35:03.1827117 :: PAREP013I Replicating Safe PSMsSessions.
22/11/2021 14:35:03.1938147 :: PAREP013I Replicating Safe PSMLiveSessions.
22/11/2021 14:35:03.7996807 :: PAREP013I Replicating Safe PSMUniversalConnectors.
22/11/2021 14:35:03.8218257 :: PAREP013I Replicating Safe PSMNotifications.
22/11/2021 14:35:03.8328487 :: PAREP013I Replicating Safe PSMUnmanagedSessionAccounts.
22/11/2021 14:35:03.8494537 :: PAREP013I Replicating Safe Transfer.
22/11/2021 14:35:03.8654577 :: PAREP013I Replicating Safe PSMPRecordings.
22/11/2021 14:35:03.8820337 :: PAREP013I Replicating Safe PSMPConf.
22/11/2021 14:35:03.8967187 :: PAREP013I Replicating Safe PSMPLiveSessions.
22/11/2021 14:35:03.9109467 :: PAREP013I Replicating Safe PSMPADBridgeConf.
22/11/2021 14:35:03.9272207 :: PAREP013I Replicating Safe PSMPADBridgeUserProfile.
22/11/2021 14:35:04.0307517 :: PAREP013I Replicating Safe PSMPADBridgeCustom.
22/11/2021 14:35:04.0371747 :: PAREP013I Replicating Safe TEST.
22/11/2021 14:35:04.0589577 :: PAREP013I Replicating Safe CyberArk-Service-Accounts.
22/11/2021 14:35:04.0682847 :: PAREP013I Replicating Safe Lin-Fin-US.
22/11/2021 14:35:04.0832497 :: PAREP013I Replicating Safe Ora-Fin-US.
22/11/2021 14:35:04.1083747 :: PAREP013I Replicating Safe Win-Srv-Fin-US.
22/11/2021 14:35:04.1223407 :: PAREP013I Replicating Safe TEST-RESTORE.
22/11/2021 14:35:04.1339357 :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
22/11/2021 14:35:04.1339727 :: GetPADRWorkingDirectory returned [C:\Program Files (x86)\PrivateArk\PADR\Conf]
22/11/2021 14:35:04.1428027 :: PADR0010I Replicate ended.
22/11/2021 14:35:05.1447417 :: PADR0099I Metadata Replication is running successfully.
  
```

Confirm Automatic Failover for PVWA and PSM

In this step we will confirm that our end users can still access critical systems via **CyberArk**.

1. Login to the **PVWA** as *John* and launch a secure connection to the target Windows machine using the account *administrator*. If everything works as expected, *John* should be able to launch the secure connection without any human intervention.



2. Lastly, login to the **PVWA** as *Mike* and navigate to **SYSTEM HEALTH**. Confirm server 10.0.10.1 once again acts as **PRIMARY** and server 10.0.10.5 acts as **DR**. Confirm all other components are connected.



Note: It may take a little longer for the **PSM for SSH** service to failover, but eventually it should failover to the functioning Vault.

3. Because we don't want to failover again accidentally, which could happen with our virtual machines during a shutdown or a startup, it is important to disable both the **DR** and the **DR_Failback** users (using either the **PVWA** or the **PrivateArk Client**) until the time you want to work again on the disaster recovery exercise.
4. You can also shut down the DR Vault if you are not using it.